

**Maximum
Temperature Data**



**Identifying
Extreme
Heatwaves**



**Delineating
Compound Events
& Heat Structures**



**Process-based
Compound
Heatwaves**

Obtain daily maximum temperature (T_{max}) for the warm season (Apr–Sep: 1940–2025) from hourly ERA5 2-m temperature over the Eastern Mediterranean & Middle East (EMME), aggregated to the 1° analysis grid



Identify local heatwave episodes from grid-specific P95th T_{max} thresholds and the persistence rule; retain regionally extensive days with $N_{\text{HW}}(t) \geq 371$ heatwave-labeled cells (the empirical regional P97.5th grid-cell-count threshold)



Group retained days into maximal consecutive-day runs, which define the compound events; then delineate the simultaneous daily heat structures within each event using event-consistent DBSCAN



Quantify unweighted PCA ellipse geometry, rigidly translate each completed ellipse to its raw-Celsius T_{max}-weighted centroid, classify event types 1–4, and analyze geometry, time, area tails, and centroid concentration